



Koneru Lakshmaiah Education Foundation

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Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I - III Semester(2025-26)
Faculty Name	Dr.Jayaram Boga
Student Name	Ramannagari Bhanu Priya
Roll Number	2520090171
Branch	CSIT
Course Outcome	1

Case Study Topic: Campus Flow – University Campus Resource Management System

Work Done Summary:

Implemented a Binary Search Tree (BST) in Java keyed by room Id / booking Id / schedule Id. Inserted multiple classroom, laboratory, faculty allocation, and student booking records into the BST. Implemented classroom scheduling, faculty allocation, laboratory booking, student service request handling, campus resource management, and infrastructure analytic operations. Deleted invalid or duplicate booking records and maintained proper BST structure. Implemented traversal methods for displaying campus schedules and resource allocations in sorted order. Added methods for searching classroom availability quickly and generating campus utilization statistics. Compared BST performance with Array List – BST provides faster searching and organized data handling for university campus management systems.

Implementation Details :

1. Created BST Node class with fields: key, room Id, facilityName, bookingDetails, left, right.

2. Insert() recursively inserts new classroom, laboratory, and booking records into the BST.
3. Search() finds classroom availability, faculty schedules, and student booking details.
4. Delete() removes cancelled or duplicate booking records.
5. InorderTraversal() displays classrooms, schedules, and bookings in sorted order.
6. ResourceAllocation() manages classroom and laboratory allocation efficiently.
7. UtilizationAnalytics() calculates campus resource usage statistics.
8. The main method demonstrated classroom booking, faculty allocation, searching, deletion, analytics, and campus schedule reporting.

Result : Screenshots/Output:

```
===== CAMPUS RESOURCE RECORDS =====
Booking ID      : 101
Room ID         : CR-201
Facility Name    : Classroom
Booking Details : Data Structures Class
-----
Booking ID      : 150
Room ID         : AUD-3
Facility Name    : Auditorium
Booking Details : Technical Seminar
-----
Booking ID      : 175
Room ID         : CR-105
Facility Name    : Classroom
Booking Details : Faculty Meeting
-----
Booking ID      : 205
Room ID         : LAB-1
Facility Name    : Computer Lab
Booking Details : Java Programming Lab
-----
Booking ID      : 250
Room ID         : LIB-2
Facility Name    : Library Hall
Booking Details : Workshop Event
-----

RESOURCE FOUND
Room ID         : LAB-1
Facility Name    : Computer Lab
Details         : Java Programming Lab

Deleting Booking ID 150...
```

```
===== UPDATED RECORDS =====
Booking ID      : 101
Room ID         : CR-201
Facility Name    : Classroom
Booking Details : Data Structures Class
-----
Booking ID      : 175
Room ID         : CR-105
Facility Name    : Classroom
Booking Details : Faculty Meeting
-----
Booking ID      : 205
Room ID         : LAB-1
Facility Name    : Computer Lab
Booking Details : Java Programming Lab
-----
Booking ID      : 250
Room ID         : LIB-2
Facility Name    : Library Hall
Booking Details : Workshop Event
-----

Total Active Resources : 4
```

Time Complexity:

- BST Insert/Delete/Search: $O(\log n)$ (average case)
- Resource Allocation Traversal: $O(n)$
- Compared to ArrayList searching $O(n)$, BST performs better for large-scale campus resource management.

GitHub Link :<https://github.com/ramannagaribhanupriya/DSA-2-PROJECT>

Student Signature	Faculty Signature